



NATIONAL INSTITUTE OF TECHNOLOGY, PATNA
MID-SEM EXAM, 2022, SESSION: 2021-22 SPRING SEMESTER

Program: B. Tech., Semester: 2nd
 Course Code: CH23101
 Full Marks: 30

Department: Civil Engg.
 Course Name: Engineering Chemistry
 Duration of Examination: 2 Hours

Answer all questions.

Q. No.	Question	Marks	CO	BL
1(a)	How is the % of S in a sample of coal determined by the Ultimate Analysis? What is the desirable amount of S? What are the adverse effects on the coal sample for high S content?	1+1+2 = 4	1	L3
1(b)	What is HCV of a fuel? Using neat diagram, explain the working of a Bomb Calorimeter. How is HCV determined by the calorimeter?	2+3+3 = 8		
1(c)	What is cracking? Explain the two industrial methods of cracking.	1 + 2 = 3		
2(a)	Why is the Equivalent Conductance of an electrolyte inversely proportional to its concentration? When is the Equivalent Conductance maximum for an electrolytic solution?	2 + 1 = 3	2	L3
2(b)	State the "Kohlrausch's Law of Independent Migration."	2		
2(c)	In the conductometric titration of a mixture of HCl and CH ₃ COOH with NaOH solution, why does ^{is} the slope of graph is initially negative and then becomes positive? Comment on the absolute value of the slope of graph in the 2 regions.	2 + 1 = 3		
2(d)	What are the advantages of using amalgam instead of metal as an electrode?	3		
2(e)	For the following redox reaction write the half-cell reactions and construct the cell. $K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 = K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$	2		
2(f)	Construct the cell for the reaction: $Co + Ni^{2+}(0.9 M) = Co^{2+}(0.11 M) + Ni$, Calculate the EMF at 298 K Given: $E^0(Co^{2+}/Co) = -0.282V$; $E^0(Ni^{2+}/Ni) = -0.236V$	2		

5.9
 46.100
 - 2.57
 43.53

LOAV
 11
 A